



[Angular upskilling program] Practical task - Epic 1 & 2

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 3 attachments (5 KB)

Angular upskilling program - example API spec.yaml; Angular upskilling program - example data model.txt; Angular upskilling program - example module structure.txt;

General information:

1. The goal of the project is to create a Kanban board application. The work is described by Epics and User Stories. They will be shared with you every two weeks.
2. Do as much as you can. This project is meant to help you on your learning path by practicing your skills in developing real-life applications.
3. All stories represent functionality. There are no strict requirements regarding UI design. However, to focus on learning Angular framework, please keep the UI simple. More complex or “fancy” designs are not the main goal of the project and are optional.
4. You are free to add your own ideas to your project, extending basic functionality covered by the tasks described in this document. Examples: drag&drop, comments on items, real time updates, user management etc.
5. Backend app – to speed up the work, it is allowed to use AI code agents **only with backend app**.
6. Code should be committed to any of the existing remote repositories.
7. Please use Conventional Commits <https://www.conventionalcommits.org/en/v1.0.0/>
8. In case tasks are not clear, please ask questions on upskilling program’s MS Teams chat or channel so that questions and answers are shared with others. If question is specific to your project (issues, bugs, review etc.) please contact directly Bartłomiej Wasiuk.
9. Data model that we propose covers all basic functionalities. You can freely extend it.

EPIC 1: Backend Persistence Service

Story 1.1 – Install and configure database

As a developer

I want to install and configure a database

So that application data can be stored persistently

Acceptance Criteria:

- DB (PostgreSQL / MySQL / MongoDB) is installed locally or via container
- Connection can be established from backend
- Basic schema is created (users, teams, items, boards)

Story 1.2 – Create backend service

As a developer

I want to create a backend application

So that it can expose REST APIs for persistence

Acceptance Criteria:

- Backend app starts successfully
- DB connection is configured
- Environment-based configuration supported

Story 1.3 – CRUD API for core entities

As a client application

I want REST endpoints for CRUD operations

So that frontend can manage data

Acceptance Criteria:

- CRUD endpoints exist for:
 - Users
 - Teams
 - Items
- REST conventions followed (GET, POST, PUT, DELETE)

EPIC 2: Authentication & User Management

Story 2.1 – User registration

As a user

I want to register an account

So that I can access the system

Acceptance Criteria:

- User can register with email/login + password
- Password is securely stored (hashed)

Story 2.2 – User login

As a user

I want to log in

So that I can use the application

Acceptance Criteria:

- Authentication mechanism implemented (session or JWT)
- Invalid credentials are rejected

Story 2.3 – User profile

As a user

I want to view my account details

So that I know my team memberships

Story 2.4 – Multiple team membership

As a user

I want to belong to multiple teams

So that I can collaborate across projects

Note: As an optional work, you can create user type enum that indicates if the user is a developer (team member) or admin. In that scenario, admin can access view with all users and the possibility to assign them or remove from any available teams. In addition, admin has the possibility to create and remove teams.

Basic version: teams are directly inserted into DB. User can do the assignment manually on the profile view.

Acceptance Criteria:

- Many-to-many relationships between users and teams
- Users can see all their teams



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